



Aerodynamic interference between road vehicles and bridge deck subjected to vortex-induced vibration

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ARTICLE INFO

Keywords:

Twin-box bridge deck
Road vehicle
Vortex-induced response
Vortex-induced force
Aerodynamic force
Wind-tunnel test

ABSTRACT

An accurate assessment of the driving comfort and safety of road vehicles moving on a long-span bridge subjected to vortex-induced vibration (VIV) is essential for bridge administrators to decide whether the bridge should be closed to traffic. However, previous assessments often ignore the aerodynamic interference between the road vehicles and the bridge deck subjected to VIV. In this study, a specific wind tunnel model is developed to explore the aerodynamic interference between road vehicles and twin-box bridge deck during VIV. The vortex-induced force (VIF) and vortex-induced response (VIR) of the twin-box bridge deck and the aerodynamic forces on the vehicle were simultaneously measured. The influence of the vehicles on the VIV of the deck was investigated, and the influence of the deck vibration on the aerodynamic forces of the vehicle was also explored. The results show that the VIR and VIF of the bridge deck were generally reduced, depending on the type, position, and number of vehicles. The aerodynamic forces of vehicles could be amplified due to the deck vibration. These findings supplement the database of vehicle aerodynamic coefficients for assessing the driving comfort and safety of road vehicles moving on a long-span bridge subjected to VIV.

Nomenclature

a	Acceleration response of the bridge deck, m/s^2
a_r	Dimensionless acceleration response of the vehicle
Amp	Mean peak displacement amplitude presented in Fig. 8, m
Amp_i	Mean peak displacement amplitude of the 9 cycles of the sub-record (i), m
A_S	Side area of the vehicle model, m^2
C_S, C_L, C_R	The side force, lift force, and rolling moment coefficients of a vehicle
D	Depth of the deck section, m
F	Natural frequency of the suspended deck section in the vertical direction, Hz
$Forc$	Mean peak force amplitude presented in Fig. 8, N/m
$Forc_i$	Mean peak force amplitude of the 9 cycles of the sub-record (i), N/m
F_S, F_L, F_R	The average values of side force, lift force and rolling moment of a vehicle, N, N, N-m
H_0	The distance from the fixed point of a vehicle to the deck surface, m
m_0, c_0	Parameters representing the non-wind induced force, kg/m , N-s/m^2
PSD	Power spectrum density
$R_{amp,i}$	The relative differences of the mean peak displacement amplitude of sub-record (i) to that shown in Fig. 8

(continued on next column)

(continued)

$R_{forc,i}$	The relative differences of the mean peak force amplitude of sub-record (i) to that shown in Fig. 8
St	Strouhal number
U	Incoming wind speed, m/s
U_r	Reduced wind speed
VIF	Vortex-induced force
VIR	Vortex-induced response
VIV	Vortex-induced vibration
y_{max}	Maximum VIV displacement amplitude of the bridge deck at a given wind speed, m
α	Wind angle of attack of the bridge deck, °
η	Non-dimensional amplitude of the bridge deck
ρ	Air density, kg/m^3

1. Introduction

As the main span of long-span bridges rises, their flexibility inevitably increases, making the bridges more susceptible to VIV (Li et al., 2014; Ge et al., 2022). Although VIV is a self-limited vibration that will not cause catastrophic collapse of the bridges, its relatively large

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<https://doi.org/10.1016/j.jweia.2024.105845>

Received 16 March 2024; Received in revised form 3 July 2024; Accepted 27 July 2024

Available online 31 July 2024

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amplitude in higher modes of vibration could make the drivers feel panicked and cause accidents. To avoid vehicle accidents, the bridge administrators tend to early close the bridge to traffic once VIV of the bridge occurs, despite that the interruption of traffic will cause economic loss and inconvenience to the public. It is therefore desirable to develop a more scientific guidance for the bridge administrators to keep the operation of the bridge as long as possible while ensuring the driving comfort and safety. To this end, an accurate assessment of the driving comfort and safety of vehicles moving on a long-span bridge, which is affected by VIV, is essential to facilitate the bridge administrators in deciding whether it should be closed to traffic or not. However, currently performed assessments of driving comfort and safety mainly concern the vehicles moving on bridges affected by buffeting, and most assessments ignore the aerodynamic interference between road vehicles and bridge deck.

To assess the driving comfort and safety of the vehicles moving on a long-span bridge affected by buffeting, the coupled vehicle-bridge system under crosswinds were established (Xu and Guo, 2004; Chen and Cai, 2004; Guo and Xu, 2006; Zhou and Chen, 2015, 2016). Guo and Xu (2006) utilized the established coupled wind-vehicle-bridge system to evaluate driving safety through different indicators. However, the aerodynamic forces and moment acting on the vehicle in their study were calculated using the aerodynamic coefficients of the vehicle on the ground, without considering that these aerodynamic forces and moment would be influenced by the bridge deck. In view of this shortcoming, Zhu et al. (2012) conducted tests in a boundary layer wind tunnel to explore the aerodynamic influence of a stationary bridge deck on the aerodynamic forces and moment of three different types of vehicles. Other scholars (Dorigatti et al., 2012; Chen et al., 2016; Kim et al., 2020; He et al., 2020; Zhang et al., 2023) also supplemented the database of aerodynamic coefficients for various kinds of vehicles on stationary bridge decks under crosswinds. They found that the aerodynamic coefficients and variations with the positions of vehicles were related to the type of bridge, wind angle of attack, and the type and size of wind barriers. However, the influence of bridge deck vibration on the aerodynamic forces acting on the vehicle under crosswinds was not considered.

To assess the driving comfort and safety of vehicles moving on long-span bridges affected by VIV, Yu et al. (2020) used the semi-empirical VIF model, proposed by Ehsan and Scanlan (1990), to simulate VIF on the bridge and then compute the dynamic responses of both the bridge and vehicle. Zhu et al. (2021) employed a sinusoidal VIF model to simulate VIF on the bridge and then explore the effects of traffic density, traffic proportion and road roughness on the driving comfort and safety. Zhang et al. (2023) utilized the semi-empirical nonlinear VIF model, proposed by Zhu et al. (2017b), to investigate the driving safety and comfort of vehicles travelling on a twin-box bridge affected by VIV with considering the multi-mode lock-in regions. However, all the VIF models used in the above studies were developed without considering the influence of vehicles on the VIF of the bridge, and the aerodynamic forces on the vehicle did not consider the effects of the bridge and/or deck vibration either.

Compared with the vehicle-bridge system affected by buffeting, it is more important to consider the aerodynamic interference between the vehicles and the bridge affected by VIV. This is because the presence of vehicles will affect the VIV amplitude of the bridge significantly, as pointed out by Zhou and Ge (2007), Yang et al. (2016), and Tan et al. (2022). However, their studies did not investigate the VIFs acting on the bridge deck and measure the aerodynamic forces of the vehicles on the deck subjected to VIV. Furthermore, the vibration of the bridge deck may also affect, to some extent, the flow field around the vehicle, thereby altering the aerodynamic forces acting on the vehicle. In addition, the higher modes of vibration and the larger amplitude of VIV of the bridge may affect the vision of drivers and cause accidents, as pointed out by Lasen et al. (2000) by analyzing the pictures of vehicles at the crest and trough of the deck during VIV.

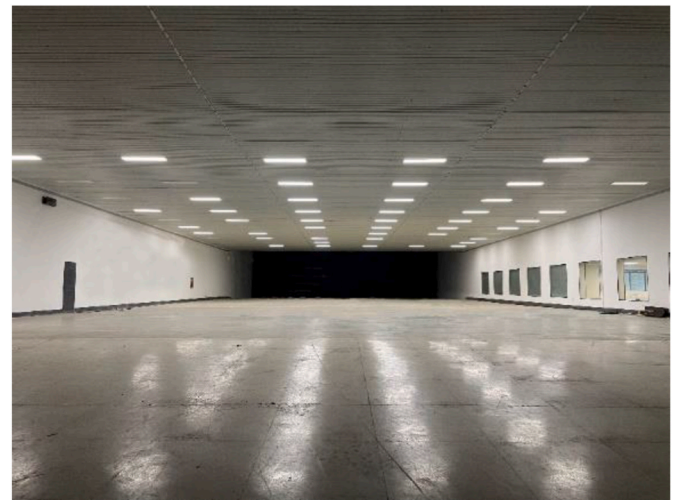


Fig. 1. The photo of XNJD-3 wind tunnel.

Therefore, towards an accurate assessment of the driving comfort and safety of road vehicles moving on a long-span bridge affected by VIV, this study investigates the aerodynamic interference between road vehicles and bridge deck during VIV. A specific wind tunnel model of a vehicle-bridge deck system subjected to VIV was developed. The VIF and VIR of the bridge deck and the aerodynamic forces on the vehicles were simultaneously measured. The influence of vehicles on the VIF and VIR of the bridge deck was investigated by placing three different kinds of vehicles on different lanes. The influence of the bridge deck and deck vibration during VIV on the aerodynamic forces of different vehicles was also explored. The remaining parts of this paper are organized as follows. Section 2 introduces the wind tunnel test arrangement including, wind tunnel, bridge deck model, vehicle models, and test cases. Section 3 discusses the test results of VIR and VIF of the bridge deck with and without the influence of vehicles with 0° wind angle of attack. Section 4 explores the interference of the deck vibration on the aerodynamic forces of the vehicle by contrasting the relevant test results with various vehicles placed on the stationary bridge deck at 0° wind angle of attack. Section 5 shows the aerodynamic interference between the vehicle and bridge deck with -3° and $+3^\circ$ wind angles of attack, while the main findings from this study are summarized in Section 6.

2. Wind-tunnel test arrangement

2.1. Wind tunnel, models, and instrumentation

To ensure VIV of the vehicle-bridge deck system being simulated appropriately in a wind tunnel, the length scale of the bridge deck and vehicle models should be as large as possible. To this end, wind tunnel tests were conducted in the XNJD-3 wind tunnel of Southwest Jiaotong University in Chengdu, China, as shown in Fig. 1. This wind tunnel is a closed-circuit wind tunnel with the length, width, and height of 36.0m, 22.5m, and 4.5 m, respectively. The smooth flow field generated in the wind tunnel can meet the test requirement that the turbulence intensity of the wind field is less than 2%.

The twin-box deck of the Xihoumen Bridge, which has a four-lane highway, was selected for this study because this bridge did suffer from VIV once. The length scale of 1:20 was used, and the geometry and dimensions of the bridge deck are shown in Fig. 2. Compared with those used by Zhu et al. (2017a), the length of the bridge deck was increased from 3.6 m to 3.9 m to accommodate the arrangement of the vehicle fleet. The blockage ratio of the bridge deck section is 3.89%, less than the traditional test threshold of 5%.

The force-measuring strips and their positions, which are illustrated in Fig. 3, were arranged to investigate the aerodynamic forces acting on

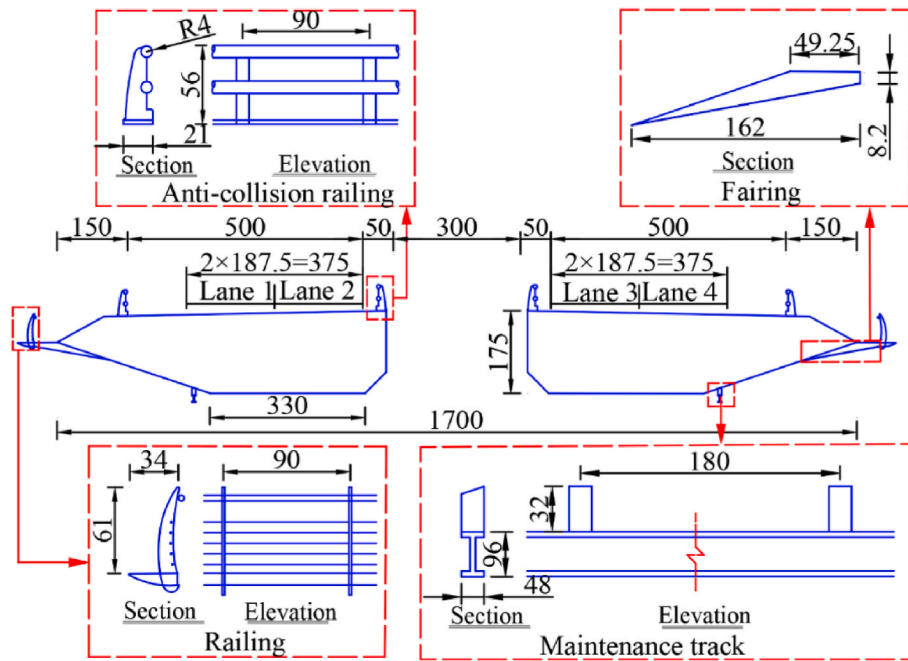


Fig. 2. Configuration of 1:20 bridge deck sectional model (unit: mm).

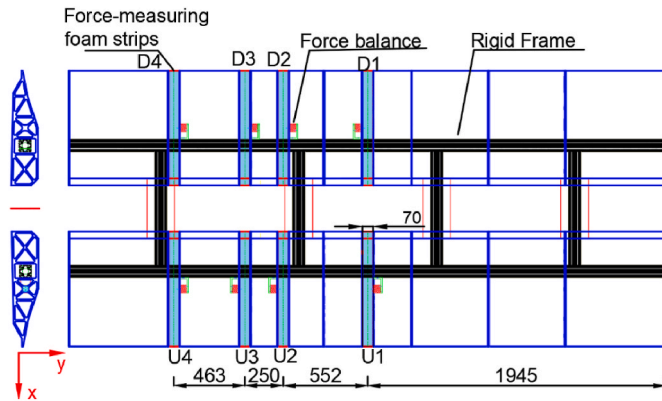


Fig. 3. Configuration of the section model (unit: mm).

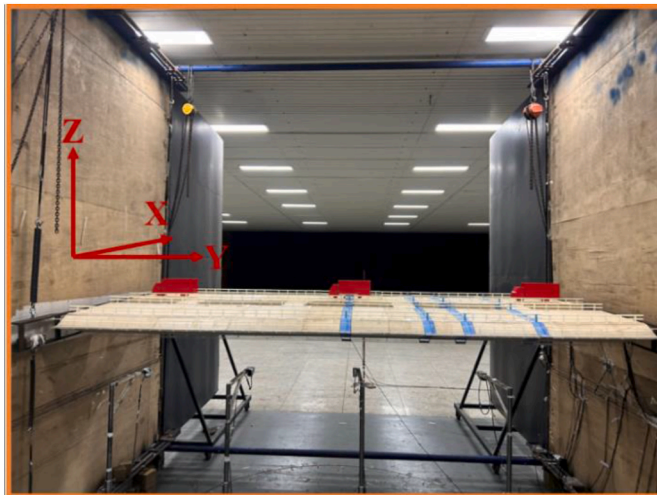


Fig. 4. Deck sectional model and its installation in wind tunnel.

the vibrating deck. The mass of each force-measuring strip was 0.11 kg. The adopted force balances were developed by Tongji University (Zhu et al., 2017a) to measure the aerodynamic forces of the bridge deck section installed in a wind tunnel. The dynamic strain gauges were embedded in the force balances with a measurement range of ± 1.2 kg in the vertical direction and a accuracy of within 4.0% F.S. (full scale). These force balances were precise enough to measure the VIF of the bridge deck section in this study. The force-measuring strip was rigidly connected to the force balance through four bolts, and then the strip-balance system was rigidly connected to the inner rigid skeleton of the bridge deck model. The natural frequency of the strip-balance system was measured as 44 Hz, which was approximately 10 times the natural frequency of the bridge deck. The assembled deck model was connected to eight springs through the rigid plate fixed on both sides of the deck model, and the eight springs were suspended on the rigid support of the fairing walls, as shown in Fig. 4. Two fairing walls were adopted to ensure the flow quality around the models and provide enough stiffness support for the vibrating deck-vehicle system. The total mass of the mounted deck model was 200.2 kg. The vertical natural frequency and damping ratio of the mounted model were 4.28 Hz and 0.38%, respectively according to the analysis of its free vibration time histories. The direction of the incoming mean wind was opposite to the positive direction of X-axis.

To accurately measure the VIR of the bridge deck, five laser displacement transducers were employed to record the displacement time histories, while one accelerometer was utilized to record the acceleration time history. To minimize the influence of the laser displacement transducers and their supports on the flow field around and the VIF on the strips, the laser displacement transducers for the U2_D2 strips and U4_D4 strips were placed on the other side of the bridge deck symmetrically to the strips U2_D2 and the strips U4_D4. The synchronous measurement of the VIF and the displacement of the bridge deck could be realized with a sampling frequency of 1024 Hz. Moreover, a micromanometer was adopted to measure the mean wind speed of the smooth flow in the wind tunnel.

Three different vehicles were selected for the subsequent tests: light car, medium truck, and bus. For each type of vehicle, there were three vehicle models: one vehicle model equipped with a force balance, named as the force-measuring vehicle, was placed on different lanes at

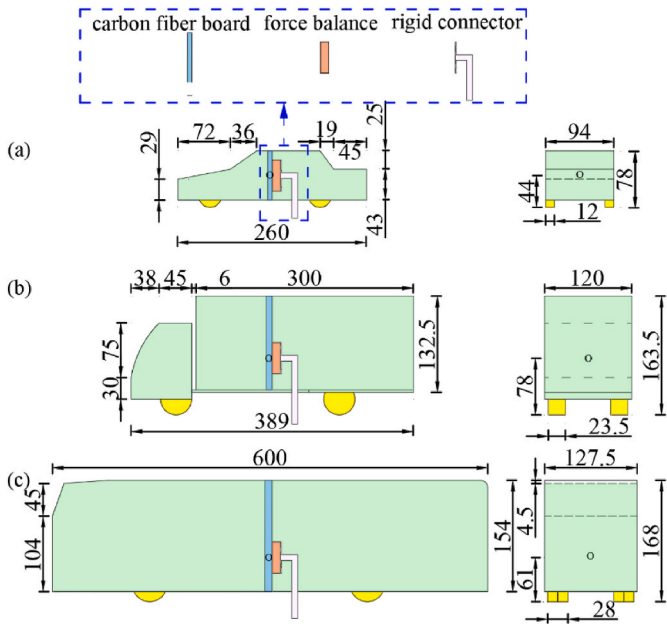


Fig. 5. Three types of vehicle models: (a) light car, (b) medium truck, and (c) bus (unit: mm).

the central-section of the bridge deck; the other two vehicles, called dummy vehicle, were fixed on the bridge deck using adhesive tape and formed a vehicle fleet together with the force-measuring vehicle on the same lane (see Fig. 4).

The geometric (length) scale of the vehicles was also set as 1:20. Fig. 5 displays the geometry and dimensions of different vehicles. To

avoid the inertia force of the vehicle exceeding the range of the force balance (a Nano43 dynamometric balance) and minimize the influence of inertia force on the measurement accuracy, the vehicle models were made of light wood with a thickness of 2 mm. The masses of the light car, medium truck and bus were 120 g, 158 g, and 234 g, respectively. Moreover, to avoid the natural frequency of the bridge deck and/or the shedding frequency of VIV being close to the natural frequency of the vehicle, several stiffening plates were installed inside the vehicles to provide sufficient stiffness. The natural frequencies of the light car, medium truck, and bus were measured as 39.95 Hz, 28.3 Hz and 24.5 Hz by applying an excitation to each vehicle and analyzing its free vibration response. Eventually, the force-measuring vehicle was rigidly connected to the force balance which was then rigidly connected to the inner rigid skeleton of the bridge deck through a rigid connector. The clearance between the tested model and the upper surface of the bridge deck is 1 mm to prevent the tire of the vehicle from contacting the bridge deck.

2.2. Test cases

There were two types of vehicle arrangements in this study: single vehicle and vehicle fleet. For the cases of single vehicle, the force-measuring vehicle was positioned at the central section of the deck in each of the four lanes. The influence range of the vehicle on the VIFs was then studied through the analysis of the VIFs measured from the four strips located in different distances from the vehicle. For the cases of vehicle fleet, two dummy vehicles were placed on the left and right sides of the force-measuring vehicle on the same lane to simulate the aerodynamic interference from other vehicles on the VIF and VIR of the bridge deck. The distance between the dummy vehicles and the force-measuring vehicle was determined by referring to the division of traffic flow in the US Highway Capacity Manual (Washington D. Highway capacity manual, 2000). As shown in Fig. 6, the distances between

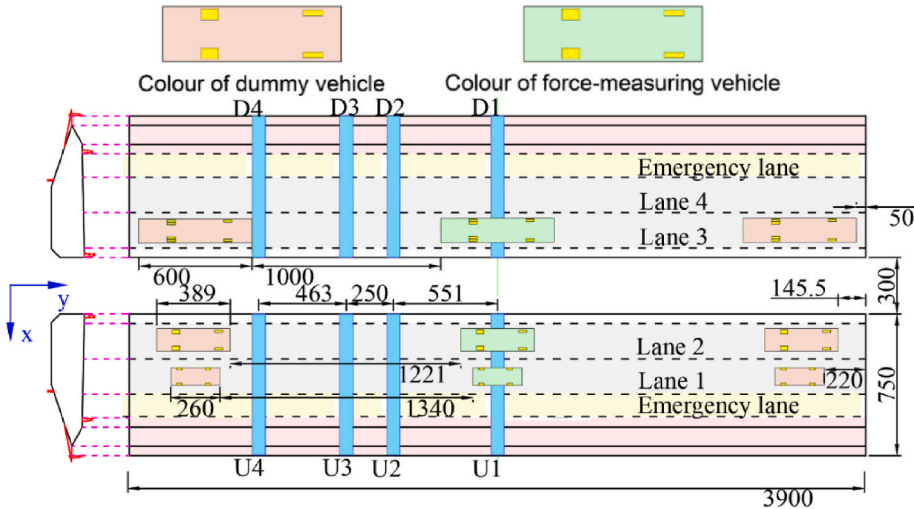


Fig. 6. The arrangement of vehicles (unit: mm).

Table 1

The test cases.

Purpose	Type of vehicle	Arrangement of vehicle	Position of vehicle	State of bridge	Wind angle of attack (°)
Investigating VIV of bridge deck	Without car	–	–	VIV	–3, 0, +3
Investigating aerodynamic forces of vehicles with fixed bridge deck	Light car, medium truck, bus	Single/Fleet	Lane 1 - 4	Stationary	0
Investigating aerodynamic interference between vehicles and deck deck	Light car, medium truck, bus	Single/Fleet	Lane 1 - 4	VIV	0
Investigating aerodynamic interference between vehicles and deck deck	Medium truck	Single/Fleet	Lane 2	VIV	–3, +3

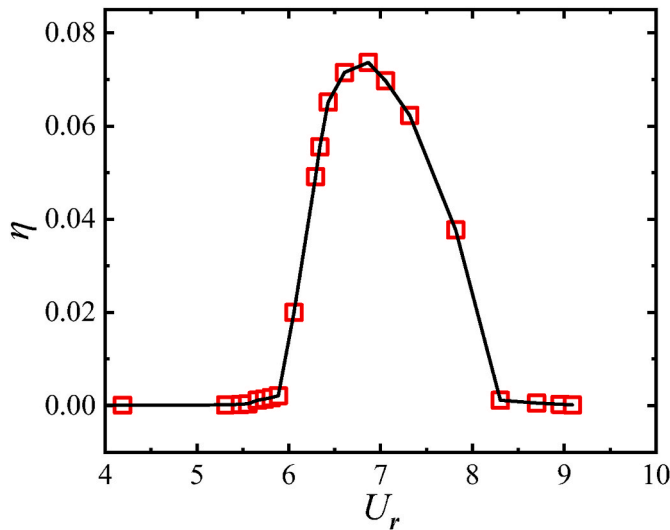


Fig. 7. VIR of bridge deck at 0° wind angle of attack.

the dummy vehicles and the force-measuring vehicle are 1.34 m, 1.221 m and 1 m for the light car, medium truck, and bus, respectively, which correspond to 26.8 m, 24.42 m, and 20 m, respectively, on site and can be classified as F level (32 vehicles/km/lane) according to the US Highway Capacity Manual. The distances between the dummy vehicles and the deck ends are shown in Fig. 6. The close location of the dummy vehicles to the fairing walls may cause some interference to the flow field at the deck ends. However, this interference is minimal compared to the end effects by the fairing walls. The effects of vehicle fleet on the VIFs were also illustrated through the analysis of the VIFs measured from the different strips with different distances from the vehicle fleet. As shown in Fig. 6, the U1_D1 strips were located at the central section of the bridge, underneath of the force-measuring vehicle directly. The U3_D3 strips were located at the middle between the force measuring vehicle and the dummy vehicle. The U4_D4 strips were located close to one of the dummy vehicles. The U2_D2 strips were located between the U1_D1 and the U3_D3 strips. The aerodynamic forces on the vehicles were recorded simultaneously with the VIF and VIR of the deck in each case.

There were totally 55 test cases in this study, and they are listed in Table 1 and summarized as follows.

- (1) One test case to study the VIV of bridge deck with 0° wind angle of attack without vehicle(s) (1 case);
- (2) Three types of vehicles were arranged in either single mode or fleet mode and placed on each of the four lanes of the stationary bridge deck in sequence ($3 \times 2 \times 4 = 24$ cases);
- (3) Three types of vehicles were arranged in either single mode or fleet mode and placed on each of the four lanes of the vibrating

bridge deck subjected to VIV in sequence to investigate the aerodynamic interference between the vibrating bridge deck and vehicles ($3 \times 2 \times 4 = 24$ cases);

- (4) Two cases to study the VIV without vehicle(s) on the deck at -3° and $+3^\circ$ wind angles of attack (2 cases);
- (5) Medium truck was placed on Lane 2 of the bridge deck at -3° and $+3^\circ$ wind angles of attack in either single mode or fleet mode to investigate the aerodynamic interference between the vibrating bridge deck and vehicles ($1 \times 1 \times 2 \times 2 = 4$ cases).

3. Influence of vehicles on VIR and VIF of bridge deck

In this section, the characteristics of VIV, such as lock-in region, VIR, and VIF, of the bridge deck without vehicles are presented. Then, the influence of the vehicles on the VIR and VIF of the bridge deck is discussed for both single vehicle and vehicle fleet cases.

3.1. VIR and VIF of bridge deck without vehicles

The vertical VIV amplitude and lock-in region of the bridge deck without vehicles at 0° wind angle of attack are shown in Fig. 7. The torsional VIV was hardly excited because the natural torsional frequency of the bridge deck is higher than 10 Hz. As illustrated by Fig. 7, the reduced wind speed of the lock-in region of the bridge deck ranges from 5.88 to 8.3. Therefore, the Strouhal number (St) was about 0.17 based on the wind speed at the start point of the VIV, and the Reynolds number of the bridge deck within the lock-in region ranged from 5.4×10^4 to 7.7×10^4 . The maximum VIV amplitude of the bridge deck appears at a reduced wind speed of 6.866. The reduced wind speed is defined as:

$$U_r = \frac{U}{fD} \quad (1)$$

where U is the incoming mean wind speed; f is the natural frequency of the suspended deck section in the vertical direction; and the D is the depth of the deck section.

The maximum non-dimensional amplitude η of the deck section is 0.07366, in which η is defined as

$$\eta = \frac{y_{max}}{D} \quad (2)$$

where y_{max} is the maximum VIV amplitude.

Fig. 8 shows the displacement response of the bridge deck at $U_r = 6.866$ and the total force acting on the U1_D1 strips. Fig. 9 shows the power spectrum density (PSD) of the VIV displacement time history at $U_r = 6.866$. It is seen that the fundamental-frequency component dominates the vibration, with smaller double and triple-frequency components present.

In the wind tunnel test, the sampling duration of each record was about 60s at one wind speed, which includes about 260 cycles. If 9 cycles are taken as a sub-record, a total of 28 sets of the sub-record could be obtained. Then, the relative differences of the mean peak displacement

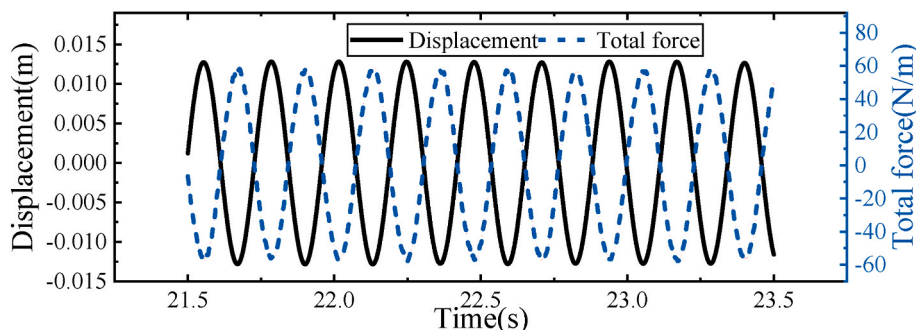


Fig. 8. The amplitude time history and total force acting on the strips U1_D1.

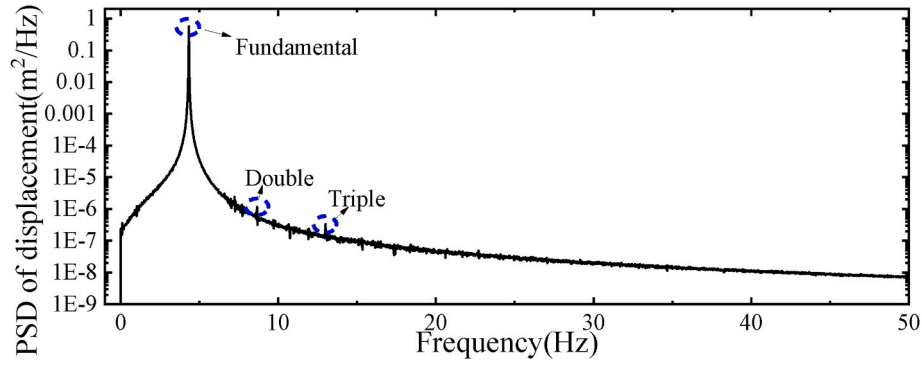


Fig. 9. PSD of VIV displacement.

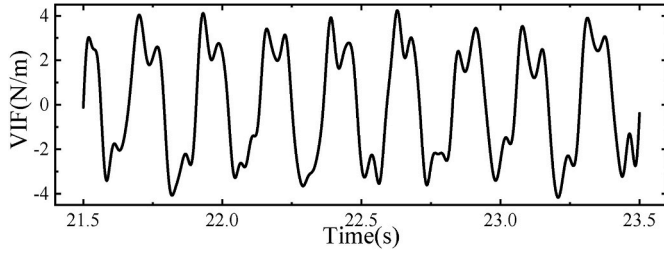


Fig. 10. Time histories of the VIF on the U1_D1 strips.

amplitude and mean peak force amplitude of each of the remaining 27 sets of the sub-record to those presented in Fig. 8 are calculated using the following equations:

$$R_{amp-i} = \frac{|Amp_i - Amp|}{Amp} \quad (3a) \quad R_{forc-i} = \frac{|Forc_i - Forc|}{Forc} \quad (3)$$

where Amp_i is the mean peak displacement amplitude of the 9 cycles of the sub-record i ; Amp is the mean peak displacement amplitude presented in Fig. 8; $Forc_i$ is the mean peak force amplitude of the 9 cycles of the sub-record i ; and $Forc$ is the mean peak force amplitude presented in Fig. 8.

The calculation results show that the maximum and mean differences were 0.364% and 0.216% for mean peak displacement amplitude and 0.524% and 0.327% for mean peak force amplitude. The corresponding standard deviations for the mean peak displacement amplitude and mean peak force amplitude were 0.0012 and 0.0015, respectively. Thus, the uncertainties of the measured vortex-induced displacements and forces are very small.

The total force shown in Fig. 8 includes the inertia force of the strip, which can be eliminated by the acceleration response of the bridge deck and the mass of the strip, the non-wind-induced force, and the VIF acting

on the strip. To extract the VIF from the total force, Zhu et al. (2013) proposed using the coefficients of m_0 , which is related to the acceleration of the bridge deck, and c_0 , which is related to the velocity of the bridge deck, to express the non-wind-induced force. Adopting their method, the coefficients of m_0 and c_0 in this study were identified to be 2.889 kg/m and 4.866 N s/m², respectively. In this way, the VIF can be extracted by subtracting the inertia force and non-wind induced force from the recorded total force.

Fig. 10 shows the time histories of the extracted VIF acting on the U1_D1 strips of the bridge deck. As illustrated, the maximum VIF extracted from the total forces is about 4 N/m. Fig. 11 shows the PSD of the identified VIF. As we can see, the fundamental frequency of VIF takes a dominant role while the double and triple frequencies of VIF are also notable. Zhu et al. (2017a) pointed out that the fundamental-frequency component contributes to vibration growth, while the triple-frequency restrains the VIV development when it is large.

3.2. VIR of bridge deck with vehicles

The VIR of the bridge deck under different reduced wind speeds with vehicles located in different lanes is shown in Fig. 12 for both single vehicle and vehicle fleet cases. The VIR of the bridge deck without any vehicles is also illustrated for comparison. From Fig. 12, we can see that compared with the lock-in region of the bridge deck without vehicle, the lock-in region tends to gradually narrow down from Lane 4 to Lane 1 with vehicles. This narrowing trend appears to be more pronounced in the vehicle fleet cases than in the single vehicle cases, and more so in the bus cases than in the light car cases. Particularly, when the bus fleet is placed on the upstream Lane 1 and Lane 2, the VIV of the bridge deck almost disappears, which indicates that the influence of the bus fleet on the VIV is significant.

Let us look at the variation of the maximum VIR amplitude ratio of the bridge deck with vehicles on different lanes, as shown in Fig. 13, in which the maximum VIR amplitude ratio is defined as

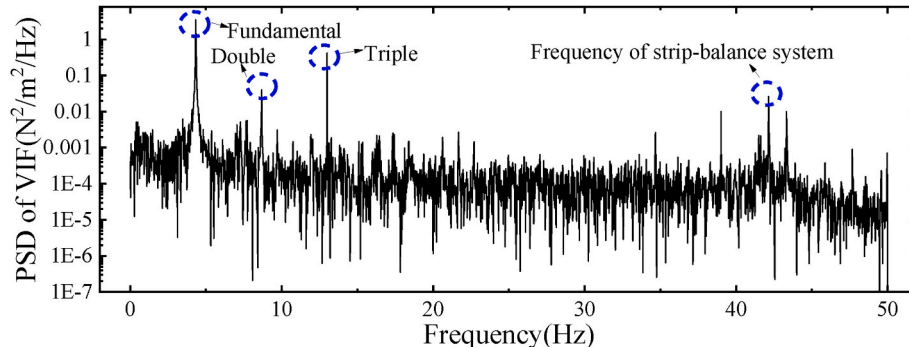


Fig. 11. PSD of VIF acting on the U1_D1 strips.

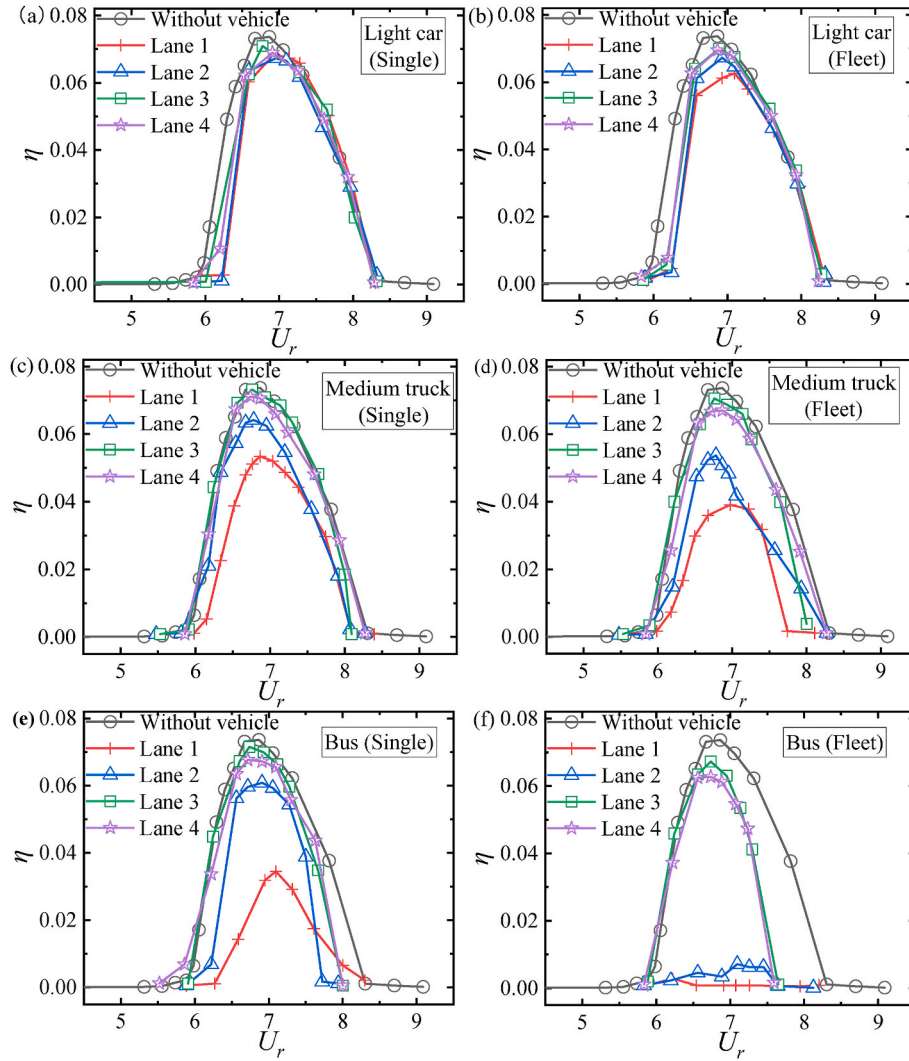


Fig. 12. VIV response of bridge deck with vehicles.

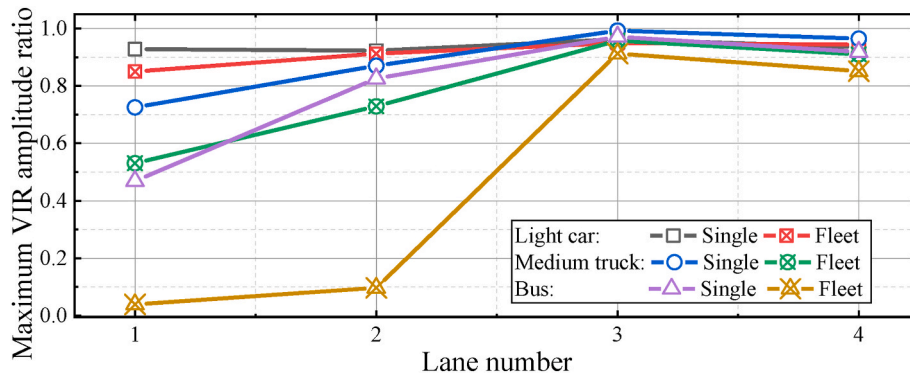


Fig. 13. Maximum VIR amplitude ratio of bridge deck with vehicles on different lanes.

$\eta_{\max}(\text{with vehicles})/\eta_{\max}(\text{without vehicles})$. For the comparison in Fig. 13, the impact of the light car on the maximum VIR amplitude of the bridge deck is the smallest among the three types of vehicles. Only when the light car fleet is placed in the upstream Lane 1, is the maximum VIR amplitude reduced by 15%, compared with that of the bridge deck without vehicles. The influence of the medium truck on the maximum VIR amplitude of the bridge deck is also small when either the single medium truck or the medium truck fleet is placed on the downstream

Lane 3 and Lane 4. However, when the single medium truck or the medium truck fleet is placed in the upstream Lane 1, the maximum VIR amplitude of the bridge deck is reduced by 27.5% and 47%, respectively, compared with that of the bridge deck without vehicles. The bus has the greatest impact on the maximum VIR amplitude of the bridge deck. When the single bus or the bus fleet is placed on the upstream Lane 1, the maximum VIR amplitude of the bridge deck is reduced by 53% and 96%, respectively, compared with that of the bridge deck without vehicles.

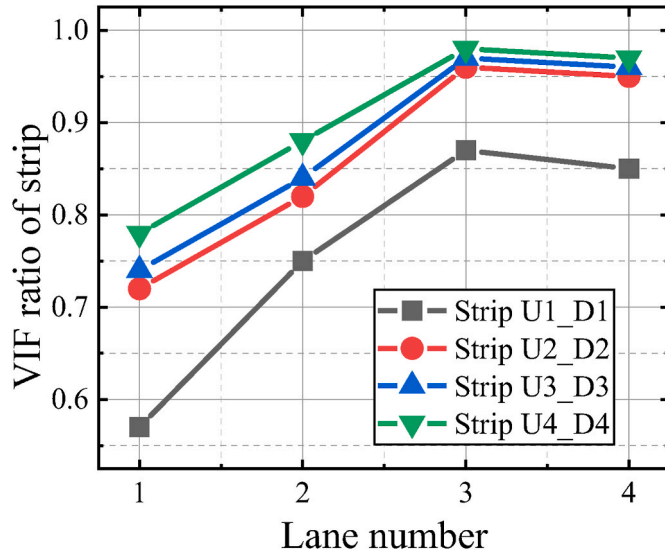


Fig. 14. Maximum VIF amplitude ratios of different strips (the single medium truck).

Nevertheless, when the single bus or the bus fleet is placed on the downstream Lane 3 and Lane 4, the influence of the vehicles on the maximum VIR amplitude of the bridge deck is less than 18%. It should be noted that even though the total length of the light car fleet (780 mm) is longer than the length of the single medium truck (389 mm) and single bus (600 mm), the maximum VIR amplitude of the bridge deck with the light car fleet is still smaller than that with the single medium truck or the single bus.

In summary, when the medium truck or bus is placed on the upstream Lane 1, in either the single vehicle case or the vehicle fleet case, the maximum VIR amplitude of the bridge deck is reduced by a large margin, compared with that without vehicles. It is probably because the medium truck or bus placed on the upstream Lane 1 hampers the formation of vortex shedding at both the leading edge and the tail of the

upstream girder. When the single vehicle or the vehicle fleet is placed on the downstream Lane 3, the vehicle has the smallest impact on the maximum amplitude. It may be because the existence of the vehicle has little influence on the development of vortices at the leading edge of the upstream girder and at the central slot of the twin-box deck. Future research using CFD to confirm these phenomena is necessary.

3.3. VIF of bridge deck with vehicles

The decrease in the maximum VIR amplitude of the bridge deck with vehicles is mainly caused by the reduction of the VIF. The existence of either the single vehicle or vehicle fleet affects the VIF to some extent, depending on the relative location of the vehicle. The impact of different vehicles on the VIF acting on the different strips of the bridge deck is investigated in this section. Only the impact results of the single medium truck and the medium truck fleet on the VIF are presented in this section because of limited space. Moreover, the influence of the vehicles on the non-wind induced forces on the strips was carefully assessed and the results showed that the maximum influence of the vehicles acting on the non-wind-induced forces on the strips was no more than 3% compared with that on the strips of the bridge deck without vehicles.

Fig. 14 presents the maximum VIF amplitude ratios of the bridge deck with the single medium truck for different strips, in which the maximum VIF amplitude ratio is defined as the ratio of the maximum VIF amplitude of the bridge deck with the single medium truck to that of the bridge deck without vehicles. The vertical axis in Fig. 14 represents the maximum VIF amplitude ratio, while the horizontal axis indicates the location of the single medium truck placed on different lanes of the bridge deck. It is seen that the maximum VIF amplitude of the stripes U1_D1 is reduced by almost 45% when the single medium truck is placed in Lane 1, compared with that without the vehicle. The impact of the single medium truck on the maximum VIF amplitude of the same strips is also significant when the medium truck is placed on other lanes. This is because the U1_D1 strips are directly sheltered by the medium truck. As the distance of the strips to the medium truck increases, the impact of the medium truck on the maximum VIF amplitude of the strips decreases. Particularly, the impact of the medium truck placed on Lane 3

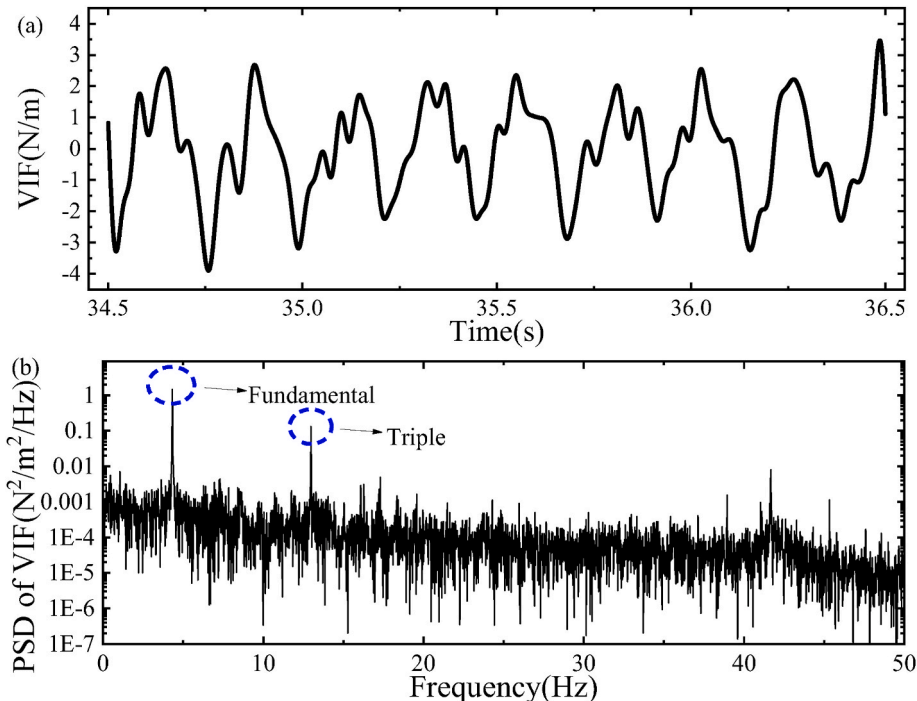


Fig. 15. VIF on the strips U1_D1: (a) time history with single medium truck placed on Lane1, (b) PSD of VIF.

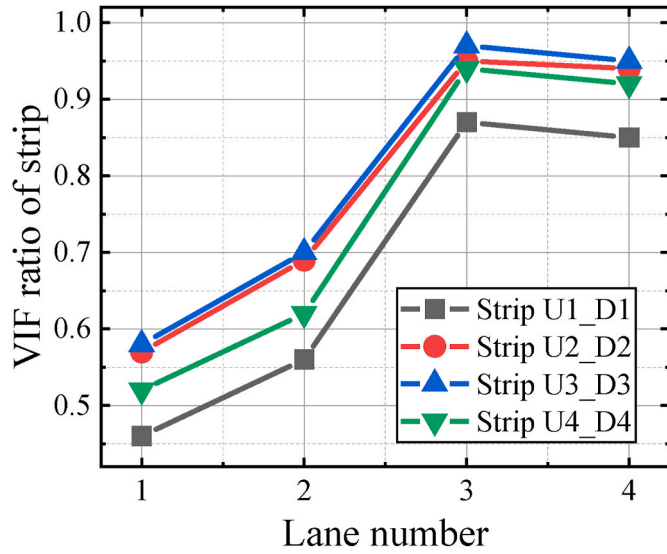


Fig. 16. Maximum VIF amplitude ratio of different strips (medium truck fleet).

and Lane 4 on the maximum VIF amplitudes of the strips U2_D2, U3_D3, and U4_D4 is small. However, the impact of the medium truck placed on Lane 1 and Lane 2 on the maximum VIF amplitudes of the strips U2_D2, U3_D3, and U4_D4 is still significant.

Fig. 15 shows the VIF time history of the strips U1_D1 and its PSD when the single medium truck is placed on Lane1. Compared with the VIF time history of the same strips without the vehicle, as shown in Fig. 10, the VIF amplitude is significantly reduced with some randomness. The comparison of the power spectra indicates that the double-component frequency in the power spectrum of VIF on the strips with the influence from the single medium vehicle almost disappears and the ratio of the spectral amplitude of the triple-component frequency to that of the fundamental frequency decreases from 11.85% to 8.56%.

The influence of the medium truck fleet on the maximum VIF amplitude of the four strips is more significant than that of the single medium truck. It is seen from Fig. 16 that the maximum VIF amplitudes

of the strips U1_D1 and strips U4_D4 are reduced by 55% and 48%, respectively, when the fleet is placed on lane 1, compared with that of the same strips without the vehicle. Only when the medium truck fleet is placed on Lane 3 and Lane 4 downstream, is the impact of the medium truck fleet on the maximum VIF amplitudes of the strips small.

Fig. 17 shows the VIF time history of the strips U1_D1 and its PSD when the medium truck fleet is placed on Lane1. The VIF amplitude of the strips is further decreased with some randomness, compared with the single medium truck case and the case without vehicle. The ratio of the spectral amplitude of the triple-component frequency to that of the fundamental frequency further decreases from 11.85% to 7.42%.

4. Influence of VIV of bridge deck on aerodynamic forces of vehicles

In this section, the influence of VIV of the bridge deck on the aerodynamic forces of a vehicle is explored. The definition of aerodynamic coefficients, which represent the corresponding aerodynamic forces and moments, of a vehicle is first introduced. Then, the aerodynamic coefficients of three different kinds of vehicles placed on the stationary bridge are presented for comparison. Lastly, the influence of the VIV of the bridge deck on the aerodynamic coefficients of the vehicles subjected to VIV of the bridge deck are explored.

4.1. Definitions of aerodynamic coefficients of vehicle

According to previous studies (Guo and Xu, 2006; Cheung and Chan, 2010), the side force (C_s), which is introduced by the directly impact of the incoming flow, the lift force (C_L), which parallel to the height direction of the vehicle, and rolling moment (C_R) are the key factors for considering the driving comfort and safety of a vehicle. Therefore, only those three forces/moment were measured and analyzed in this study. The aerodynamic coefficients of vehicle can be expressed as follows:

$$C_s = \frac{F_s}{0.5\rho U^2 A_s} \quad (4)$$

$$C_L = \frac{F_L}{0.5\rho U^2 A_s} \quad (5)$$

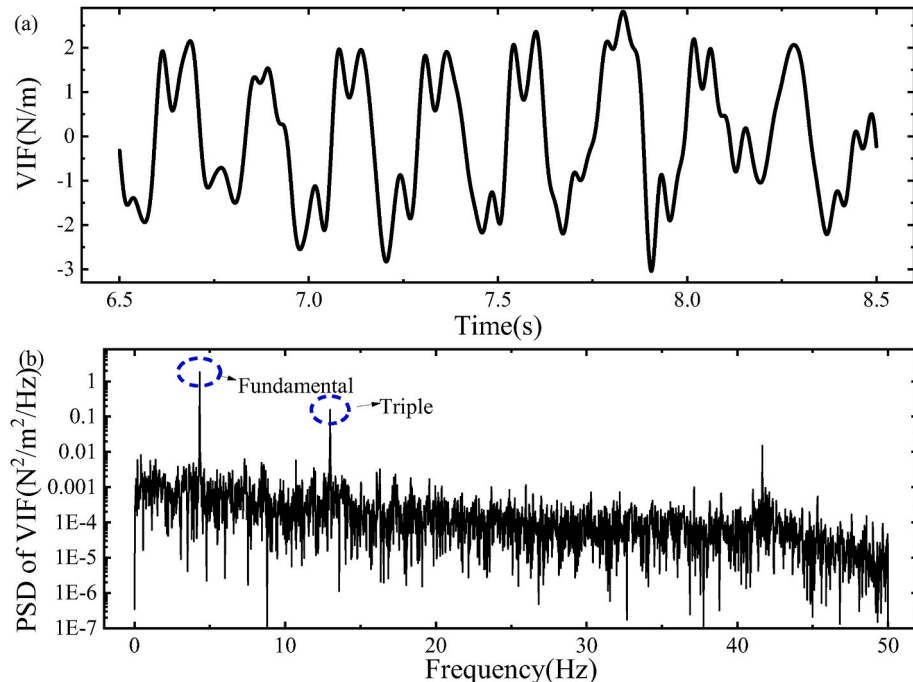


Fig. 17. VIF on the strips U1_D1: (a) time history with medium truck fleet placed on Lane 1, (b) PSD of VIF.

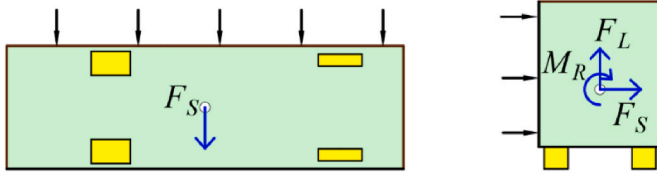


Fig. 18. Sign conventions of aerodynamic forces and moment.

$$C_R = \frac{M_R}{0.5\rho U^2 H_0 A_s} \quad (6)$$

where F_s , M_R and F_L are the average values of side force, rolling moment and lift force recorded by the force balance, respectively; ρ is the local air density ($=1.104 \text{ kg/m}^3$), calculated based on the test conditions in wind tunnel; U is the incoming mean wind speed; A_s is the side area of the vehicle model; and H_0 is the distance from the fixed point (O) to the deck surface as illustrated in Fig. 5. Fig. 18 displays the direction of the forces and moment relative to the vehicle model.

4.2. Aerodynamic coefficients of vehicles on stationary bridge deck

For this case, the bridge deck was fixed by replacing the eight springs with eight rigid rods to ensure that the vibration of the bridge deck did not occur during tests. The incoming mean wind speed for this case was 7.15 m/s . Taking the height of vehicles as the feature size, the Reynolds numbers of the light car, medium truck and bus are about 4×10^4 , 8.25

$\times 10^4$, and 8.53×10^4 at this wind speed. Fig. 19 shows the aerodynamic forces and moment coefficients of vehicles on the deck with 0° wind angle of attack.

As illustrated in Fig. 19, the aerodynamic coefficients and their variations with the position of the single light car are almost the same as those of the light car fleet, indicating that the two dummy light cars do not affect the aerodynamic forces and moment on the force-measuring light car. Compared with those of the medium truck and the bus, not only the lane position of the light car has small influence on its side and lift force coefficients, but also the aerodynamic coefficients of the light car on any lanes of the bridge deck are relatively small. It also can be seen that the vehicle position of the medium truck and bus has a similar influence on the aerodynamic coefficients. The influence of vehicle numbers on the aerodynamic coefficients is small that the coefficients of the single medium truck are almost the same as those of the medium truck fleet. However, the aerodynamic coefficients of the single bus are smaller than those of the bus fleet, except for the lift coefficient. For this twin-box bridge deck, all the aerodynamic coefficients of vehicles reach the peak values at Lane 3 except for the lift coefficient of the light car, in which the maximum value occurs at Lane 2. This is because when placed on the downstream Lane 3, the vehicle has a minimal impact on the development of vortices at the leading edge of the upstream deck and at the central slot of this twin-box deck.

4.3. Aerodynamic coefficients of vehicles with VIV of bridge deck

The aerodynamic forces and moments acting on the three types of vehicles were measured, and the aerodynamic coefficients were

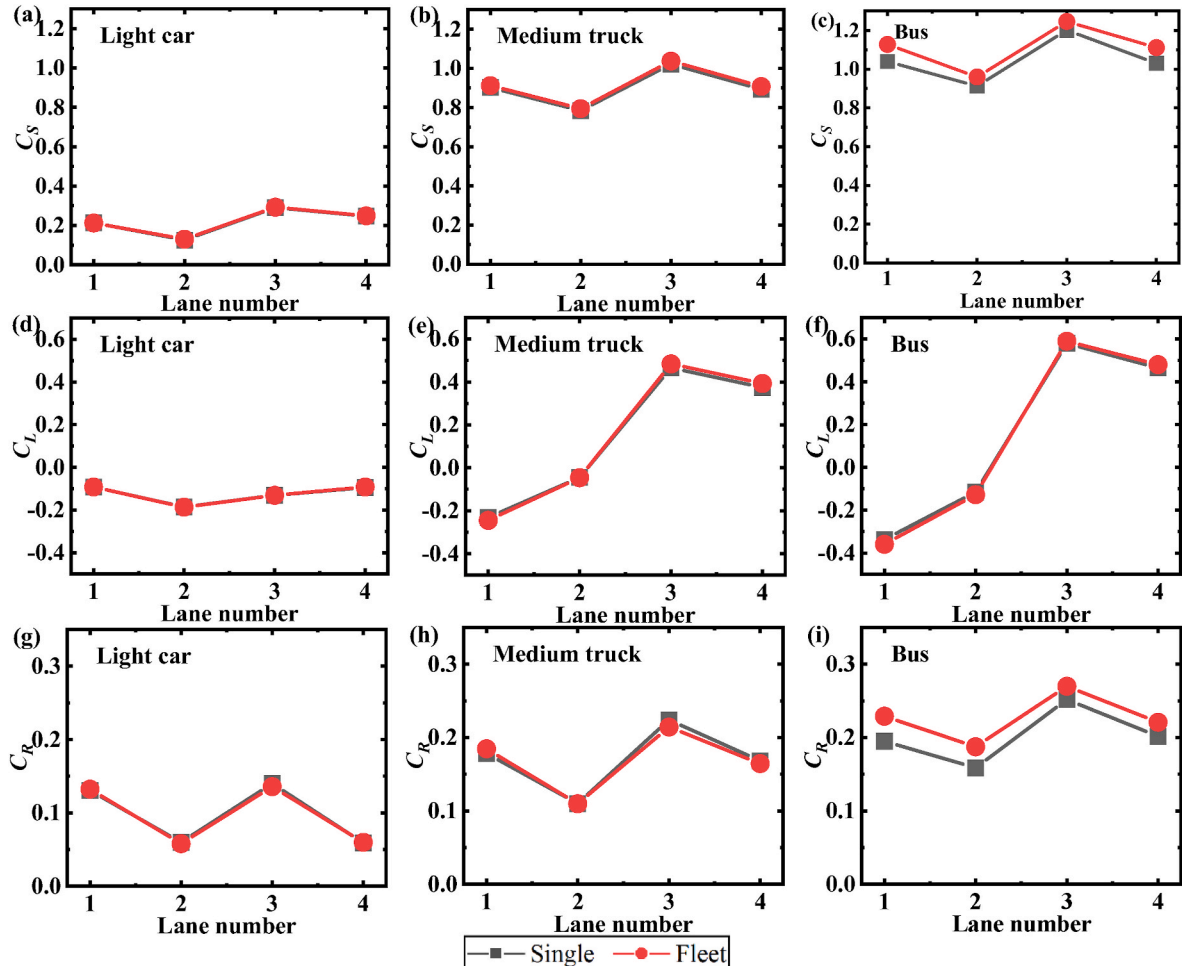


Fig. 19. Aerodynamic coefficients of different vehicles on stationary bridge deck.

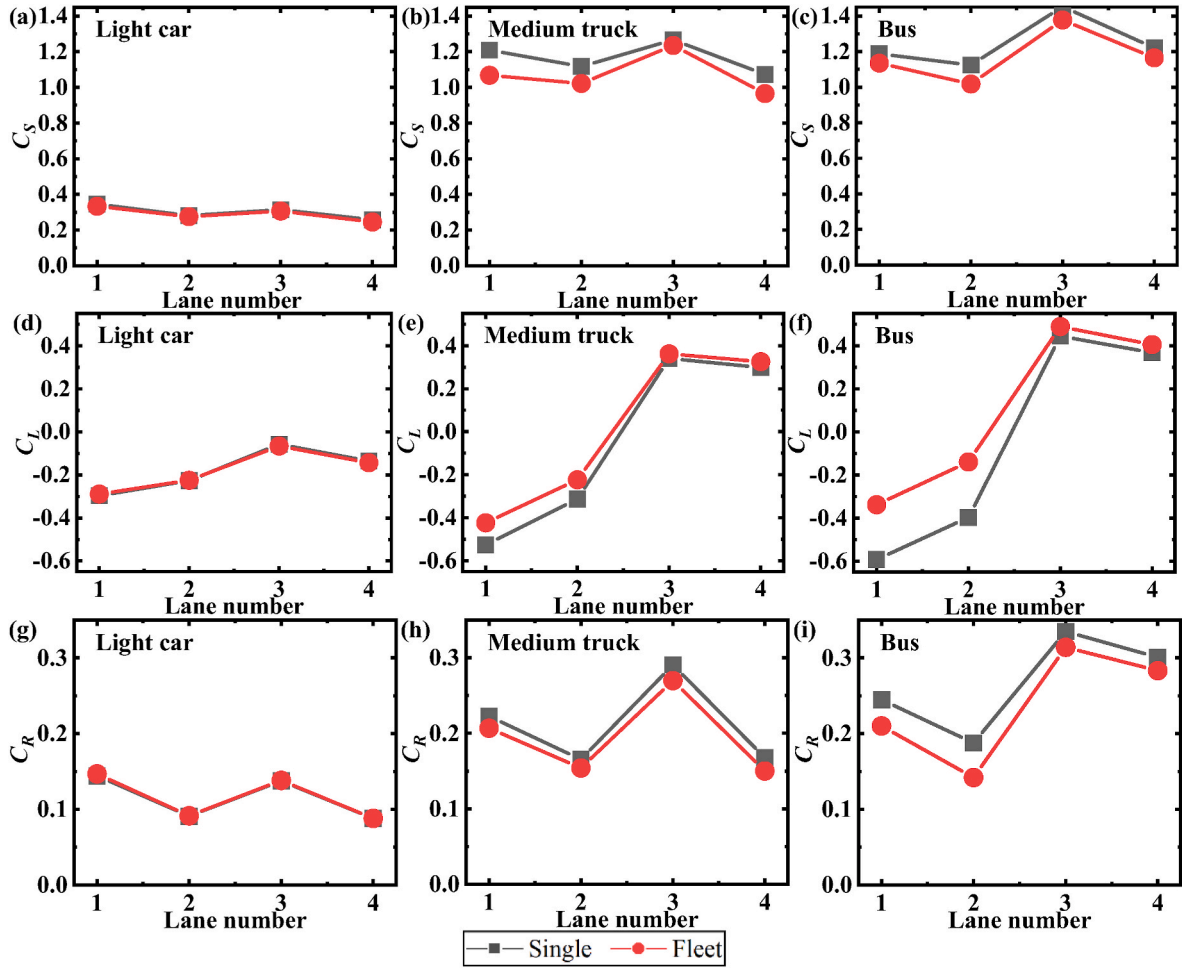


Fig. 20. Aerodynamic coefficients of three kinds of vehicles with VIV of bridge deck.

calculated accordingly and are shown in Fig. 20. As illustrated, the aerodynamic coefficients of the single light car are almost the same as those of the light car fleet. The side force and moment coefficients of single medium truck and single bus are, however, respectively larger than those for truck fleet and bus fleet, but the lift force coefficients for the single truck and single bus are, respectively, smaller than those of the medium truck fleet and bus fleet.

Fig. 21 shows the variation of the aerodynamic coefficient ratio with vehicles on different lanes, in which the aerodynamic force coefficient ratio is defined as the ratio of the aerodynamic coefficient of a vehicle on the vibrating deck to that measured from the vehicle on the stationary deck. As can be seen from Fig. 21(a), because of the VIV of the deck, the C_s ratios are much larger than one for all vehicle cases. The maximum C_s ratio for each type of vehicle appears when the vehicles are placed on Lane 2, except for the bus fleet, in which the VIV almost disappears when the bus fleet is placed on Lane 1 and Lane 2. The maximum C_s ratio among the three types of vehicles is 1.53 for the light car. The maximum C_s ratios for the single medium truck and medium truck fleet are, respectively, 1.42 and 1.29 when the medium truck/fleet is placed on Lane 2. The maximum C_s ratios for the single bus and the bus fleet are 1.29 and 1.13 when they are placed on Lane 2 and Lane 4, respectively. The C_L ratio is also significantly influenced by the vibrating deck section, as illustrated in Fig. 21(b). The maximum C_L ratio for the single light car, single medium truck, and single bus are 2.97, 6.89, and 3.44, respectively, and the maximum C_L ratio for the light car fleet, medium truck fleet, and bus fleet are 2.88, 4.86, and 1.08, respectively. It is worth noting that when vehicles are placed on Lane 1 and Lane 2, all the C_L ratios are larger than one, but when vehicles are placed on Lane 3 and

Lane 4, all C_L ratios are slightly less than one. As shown in Fig. 21(c), the variation of the C_R ratios of vehicles with the lane is, in general, similar to that of the C_s ratio. The maximum C_R ratios for the single light car, single medium truck, and single bus are 1.52, 1.37, and 1.31, respectively, while they are 1.57, 1.32, and 1.16 for the light car fleet, medium truck fleet and bus fleet. Clearly, the amplification of aerodynamic coefficients of vehicle is significant because of the VIV of the bridge deck.

4.4. Acceleration responses of vehicles with VIV of bridge deck

The vertical acceleration response of the vehicle is another important factor affecting the driving comfort. As the vehicles were rigidly connected to the bridge deck, the acceleration response of the bridge deck during VIV is the same as that of the vehicle during VIV in this study. The maximum dimensionless acceleration response of the vehicle on various lanes is shown in Fig. 22 and its definition is given as:

$$a_r = a/f^2 D \quad (7)$$

where f , D , and a are the natural frequency of the bridge deck in the vertical direction, the depth of the bridge deck, and the maximum acceleration response of the bridge deck, respectively.

As shown in Fig. 22, the variation of the maximum dimensionless acceleration response of the vehicle with lane position is similar to that of the maximum VIV amplitude ratio of the bridge deck, as shown in Fig. 13. The peak value of the dimensionless acceleration response for all vehicle cases appears when they are placed on Lane 3. The peak value of

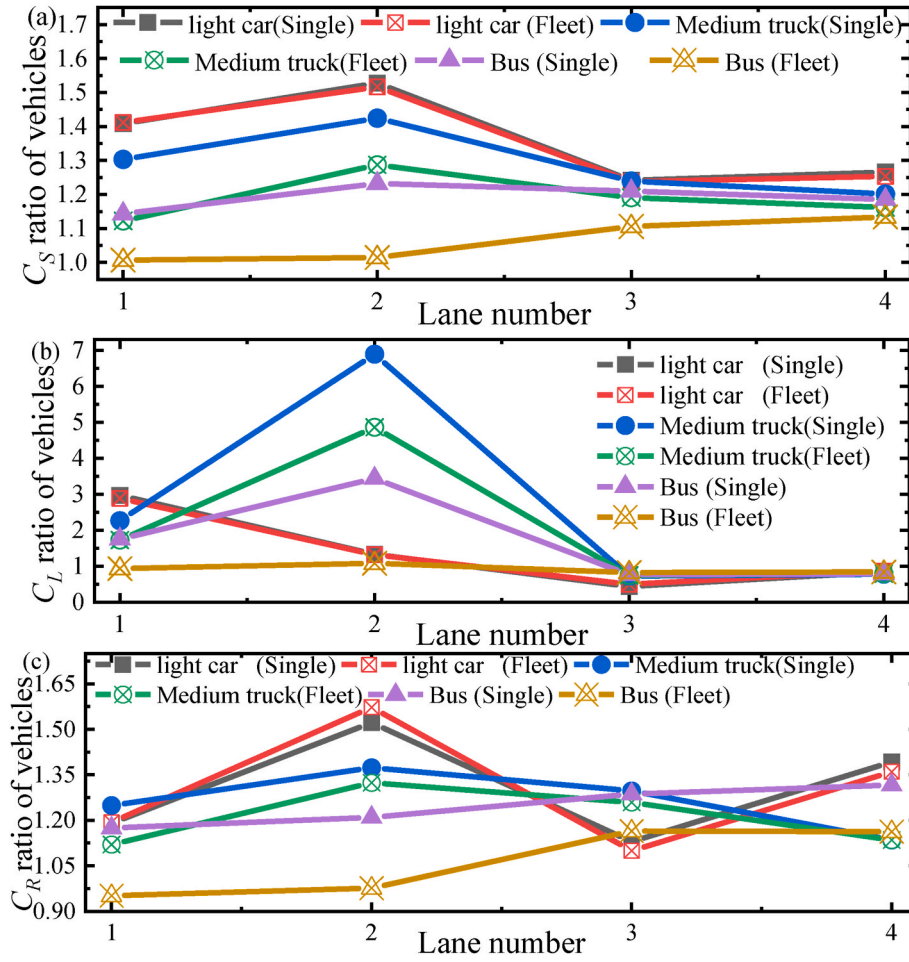


Fig. 21. Variation of aerodynamic coefficient ratios of vehicles with lane.

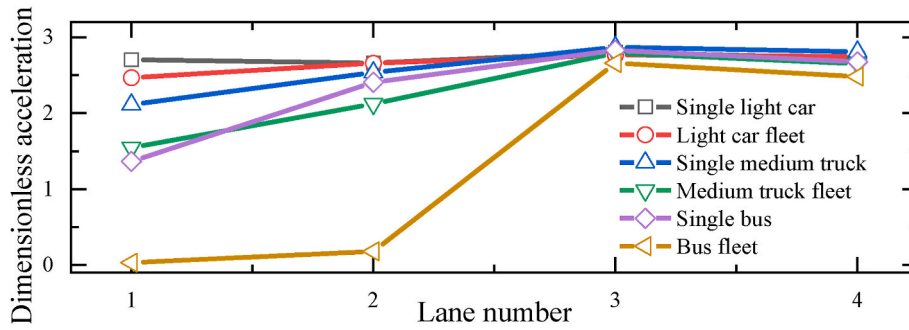


Fig. 22. The maximum dimensionless acceleration responses of vehicles at different lanes.

the dimensionless acceleration response for the single light car, single medium truck, and single bus are 2.83, 2.87, and 2.81, respectively, while the corresponding values for the light car fleet, medium truck fleet, and bus fleet are 2.77, 2.79, and 2.66, respectively.

5. Effect of wind angle of attack on aerodynamic interference

It is commonly known that the VIR of the bridge deck is extremely sensitive to its wind angle of attack. Therefore, -3° and $+3^\circ$, were selected, and the bridge deck with and without vehicles at the two wind angles of attack was tested again so that the effect of wind angle of attack on the aerodynamic interference between the bridge deck and vehicles can be analyzed. The VIR and VIF of the bridge deck at -3° and $+3^\circ$

without vehicles is first investigated and compared with those at 0° wind angle of attack in this section. Then, the VIR and VIF of the bridge deck and the aerodynamic coefficients of the single truck and the truck fleet placed on Lane 2 at various wind angles of attack are discussed.

5.1. VIR and VIF of bridge deck without vehicles at different wind angles of attack

The VIR amplitudes and lock-in regions of the bridge deck without vehicles at 0° , -3° , and $+3^\circ$ wind angles of attack are displayed in Fig. 23. It can be seen that the VIR amplitude and lock-in region of the bridge deck at -3° are similar to those of the bridge deck at 0° . The maximum VIR amplitude of the bridge deck at $+3^\circ$ is, however, much

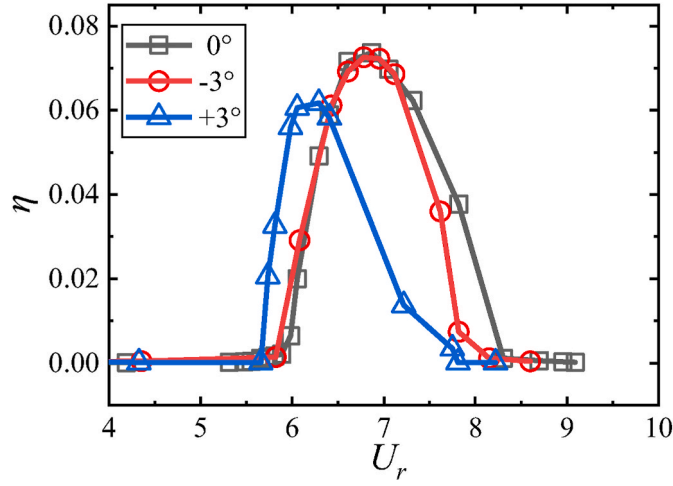


Fig. 23. VIR and lock-in regions of bridge deck without vehicles at different wind angles of attack.

smaller than that at 0° and -3° . This indicates that the bridge deck concerned in this study is more sensitive to VIV when its wind angle of attack ranges from 0° to -3° than from 0° to $+3^\circ$.

Fig. 24 shows the VIFs and power spectra of the bridge deck with -3° and $+3^\circ$ wind angles of attack without vehicles. As illustrated, the amplitude and dominant frequency components of VIF at -3° wind angle of attack are similar to those at 0° . At $+3^\circ$ wind angle of attack, the amplitude of VIF is reduced and the double-frequency component becomes more prominent than the triple-frequency component.

5.2. VIR and VIF of bridge deck with medium truck at different wind angles of attack

The VIR characteristics of the bridge deck with the single medium truck and the medium truck fleet placed on Lane 2 at -3° , and $+3^\circ$ wind angles of attack are shown in Fig. 25, together with those of the bridge deck without vehicles. It shows that with the medium truck placed on Lane 2, the maximum VIR amplitude of the bridge deck is much smaller than that of the bridge deck without vehicles. This is particularly true for the case of the medium truck fleet placed on Lane 2. The maximum VIR amplitude of the bridge deck with the medium truck fleet at $+3^\circ$ wind

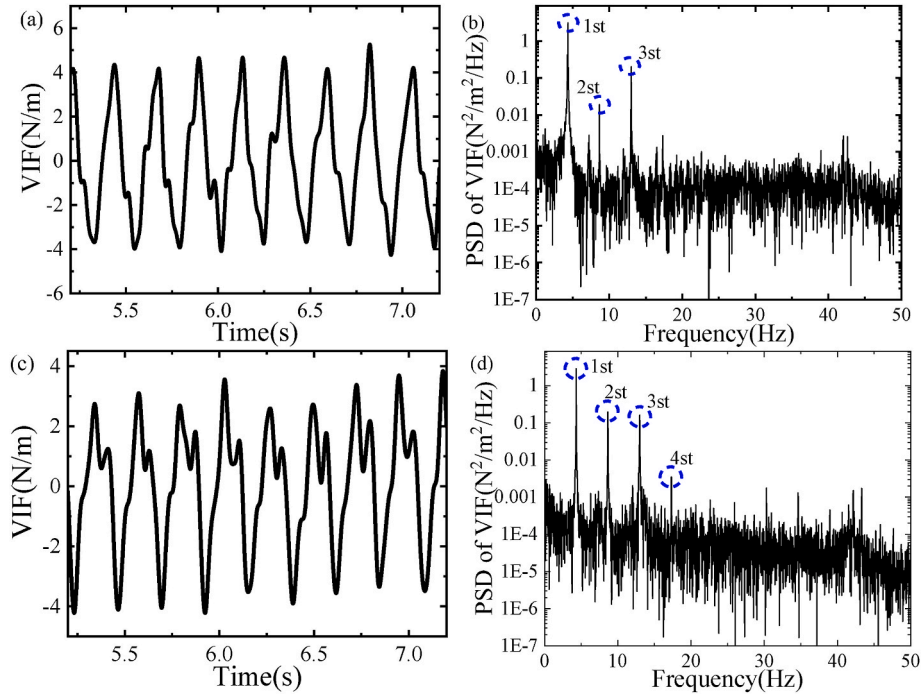


Fig. 24. VIFs and power spectra of bridge deck: (a, b) at -3° , and (c, d) at $+3^\circ$.

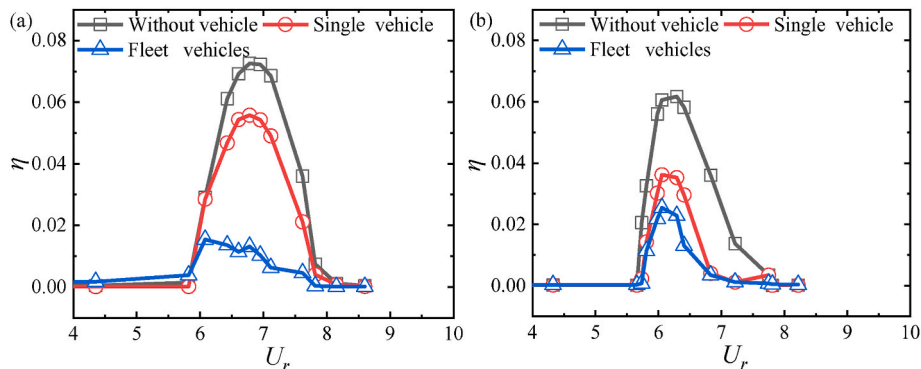


Fig. 25. VIR amplitudes and lock-in regions of deck with medium truck: (a) -3° , (b) $+3^\circ$.

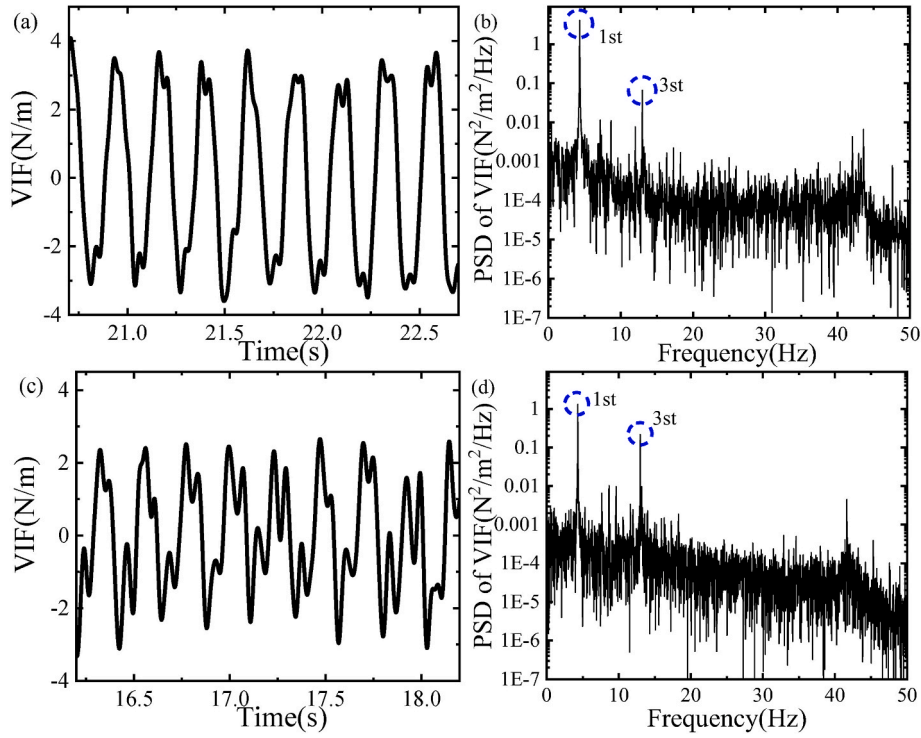


Fig. 26. VIFs and power spectra of U1_D1 strips with single medium truck: (a, b) at -3° , and (c, d) at $+3^\circ$.

angle of attack is reduced by two-thirds, compared with that without vehicle at the same wind angle of attack. The maximum dimensionless VIR amplitude of the bridge deck with the medium truck fleet at -3° wind angle of attack is only 0.015, compared with 0.075 that without vehicle at the same wind angle of attack. The effect of wind angle of attack on the aerodynamic interference between the bridge deck and vehicle is significant.

Fig. 26 shows the VIF time histories and PSD of the U1_D1 strips, which were directly sheltered by the medium truck at -3° and $+3^\circ$ wind angles of attack. As the VIF of the bridge deck with the medium truck fleet placed on Lane 2 is quite small, the corresponding results for this case are not presented here. It can be seen that the VIF amplitude of the bridge deck with the single medium truck placed on Lane 2 at $+3^\circ$ wind angle of attack is much smaller than that at -3° wind angle of attack. Moreover, the double-frequency components of VIFs in both cases disappear, compared with that without vehicle. Clearly, the effect of wind angle of attack on the aerodynamic interference between the bridge deck and vehicle is significant.

5.3. Effect of wind angles on attack on the aerodynamic coefficients of medium truck

The aerodynamic coefficients of the medium truck fleet placed on Lane 2 at different wind angles of attack are almost the same as those of the single medium truck placed on Lane 2. Therefore, only the aerodynamic coefficients of the single medium truck placed on Lane 2 at different wind angles of attack are shown in Fig. 27 (α is the wind angle of attack of the bridge deck) for both the stationary deck and the deck with VIV. It can be seen that with VIV of the bridge deck, both the side force coefficients and lift force coefficients increase, compared with the corresponding aerodynamic forces with vehicles on the stationary deck. However, the VIF does not affect the rolling moment coefficient too much. Compared with those under the stationary deck, the side force and lift force coefficients are more sensitive to the wind angle of attack in the cases of VIV of the bridge deck. The maximum aerodynamic coefficients of the single medium truck placed on Lane 2 occur in the case of VIV of the bridge deck at 0° .

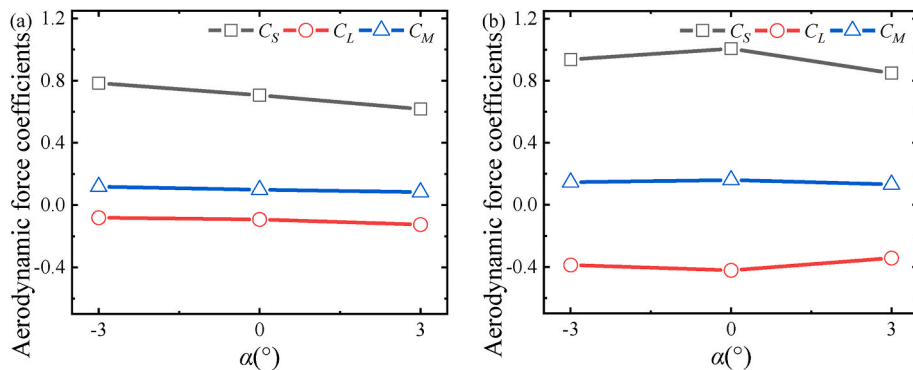


Fig. 27. Aerodynamic coefficients of single medium truck placed on Lane 2 at different wind angles of attack: (a) stationary bridge deck, and (b) bridge deck with VIV.

6. Conclusions

A specific wind tunnel model of a vehicle-bridge deck system subjected to VIV has been developed in this study to explore the aerodynamic interference between road vehicles and bridge deck during VIV. The main findings and conclusions from this study can be summarized as follows.

1. The VIR is extremely sensitive to the vehicles, and the sensitivity depends not only on the position but also on the type and number of the vehicle on the bridge deck. When placed on the upstream lanes of the bridge deck, the VIR amplitude of the bridge deck is reduced significantly, especially when it is placed on Lane 1. Nevertheless, the influence of the vehicle on the VIR amplitude of the deck is small when it is placed on the downstream lanes of bridge deck.
2. The influence of vehicles on the VIF of strips of the bridge deck relies on the distance of strips to the vehicle and the position of the vehicle on the deck. When the single medium truck is placed on Lane 1, the VIF amplitude is reduced by 45% for the strips U1_D1 as it is directly sheltered while the VIF amplitude is reduced by 22% for the strip U4_D4, which is the farthest from the vehicle. When the single medium truck is placed on the downstream lanes of the bridge deck, the influence of the vehicle on the VIF of stripes is small.
3. The aerodynamic coefficients of the vehicle are related to its type and position. When the bridge deck is stationary, the bus seems to be a critical vehicle as all the aerodynamic coefficients of the bus are all larger than those of the light car and the medium truck. Besides, Lane 3 tends to be the most dangerous lane where all the aerodynamic force coefficients of the vehicle reach their peak values except for the C_L coefficient of the light car.
4. The aerodynamic coefficients are significantly affected by the vibration of the bridge deck. The maximum C_S coefficient ratio of the light car, medium truck, and bus are 1.52, 1.42, and 1.23, respectively. The maximum C_R coefficient ratio of the light car, medium truck, and bus are 1.56, 1.37 and 1.31, respectively. When the single medium truck is placed on Lane 2, the C_L coefficient of the medium truck is increased by 6.89 times compared with that on the stationary deck.
5. The influence of the vehicle on VIV of the bridge deck at -3° and $+3^\circ$ is similar to that at 0° wind angle of attack. The VIR amplitude of the bridge deck is dramatically reduced especially for vehicle fleet cases. Besides, the amplitude of VIF is also cut down and the frequency components of VIF are changed as well.
6. The side force and lift force coefficients are more sensitive to the wind angle of attack in the cases of VIV of the bridge deck than in those cases of the stationary. The maximum aerodynamic coefficients of the single medium truck placed on Lane 2 occur in the case of VIV of the deck at 0° .

CRedit authorship contribution statement

Hao-Yang Li: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **You-Lin Xu:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Le-Dong Zhu:** Supervision, Methodology, Investigation, Data curation. **Guo-Qing Zhang:** Writing – original draft, Software, Investigation. **Bo-Man Cheng:** Software, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The works described in this paper are financially supported by the Changjiang Scholars Program of the Ministry of Education of China (SWJTU-WB0100111012209) to which the authors are most grateful. Any opinions and conclusions presented in this paper are entirely those of the authors.

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